**7** The count rate $C$ of a mixture of two radioactive nuclides X and Y is measured and the variation with time $t$ of ln $C$ is plotted and shown in Fig. 7.1. The readings show the count rate after the background count rate has been subtracted from it.

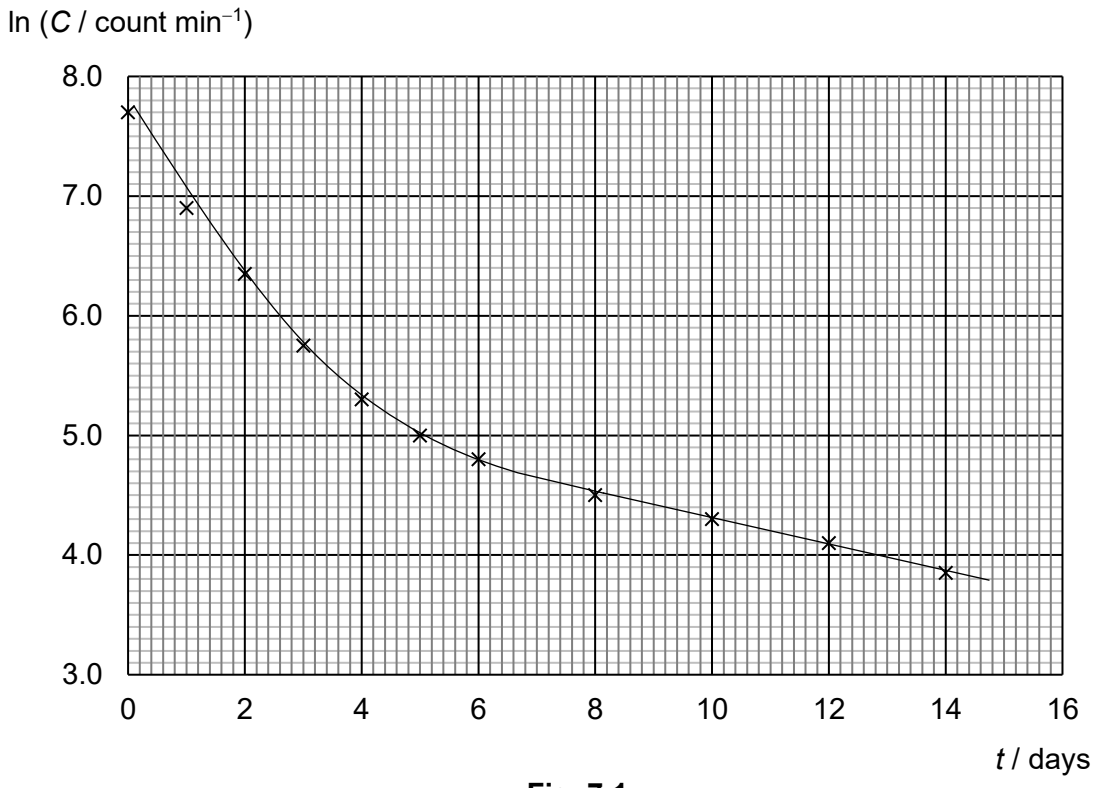


**Fig. 7.1**

It is known that the half-life of X is much longer than that of Y. The graph approaches a straight line eventually.

**(a)** By using an equation with the symbols defined below, explain why the graph approaches a straight line eventually.

Given $C_X$ : count rate of X at time $t$,
$C_{X_0}$ : initial count rate of X,
$\lambda_X$ : decay constant of X.

.....

.....

.....

.....

..... [3]

**(b)** **(i)** Determine the initial count rate of X and its half-life using information from Fig. 7.1.

initial count rate = ..... count $min^{-1}$

half-life = ..... days [5]

**(ii)** Estimate the count rate of the mixture at $t$ = 19 days.

count rate = ..... count $min^{-1}$ [2]

**(c)** One of the radioactive nuclides in the mixture can be used as a tracer to check whether the thyroid gland is absorbing iodine normally from the blood in a human body. The tracer nuclei decay by emitting beta particles and gamma rays, and the gamma rays are monitored from outside the body close to the thyroid using a detector.

A dose of iodine with the tracer is injected into a patient's blood, and 20% of that dose is absorbed by the thyroid gland. The detector records 0.40% of the gamma rays emitted.

**(i)** Suggest a reason why the beta particles are not monitored.

.....

..... [1]

**(ii)** Suggest a reason why the detector records only 0.40% of the gamma rays emitted from the radioactive nuclei inside the thyroid.

.....

..... [1]

**(iii)** In administering radioactive nuclides for medical applications, factors such as the half-life of the source must be considered.

Suggest two other significant factors that should be considered.

.....

.....

.....

.....

..... [2]

**(d)** Source Y has a half-life of 20 hours. In the absence of source Y, a constant average count-rate of 15 $s^{-1}$ is recorded by a radiation detector.

Immediately after source Y is placed 30 cm from the detector, the average count-rate rises to 100 $s^{-1}$.

Determine the distance the detector should be placed from source Y in order to detect the same average count-rate of 100 $s^{-1}$ 60 hours later.

Assume that source Y is a point source emitting radiation in all direction.

(Sample X is not present in this part of the experiment.)

distance = ..... cm [4]

**(e)** Fig. 7.2 shows the variation with time of the cumulative counts (total count) of a small sample of X and of the background radiation.

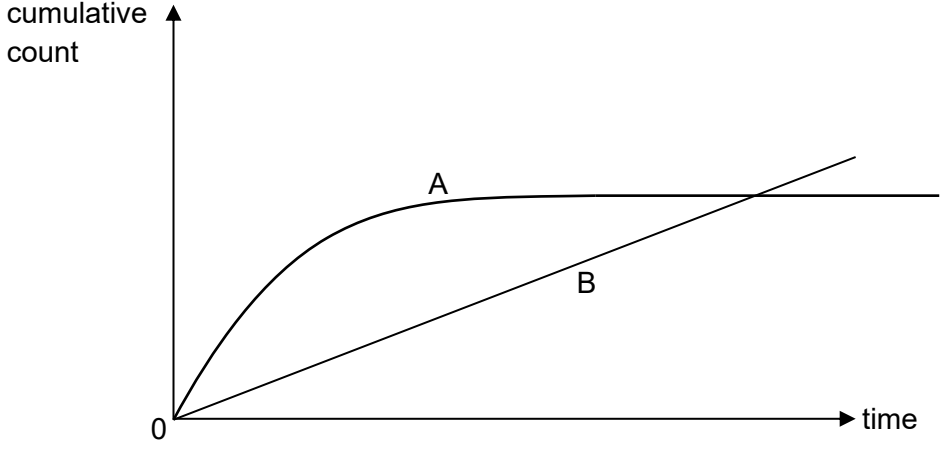


**Fig. 7.2**

**(i)** State and explain which graph, A or B, in Fig. 7.2 represents the cumulative count of the background radiation.

.....

..... [1]

**(ii)** Sketch and label on Fig. 7.2, a graph that represents the cumulative count of a sample of Y that has half the initial number of nuclei as that of sample X.

[1]